Extracting the First Prime Greater than m

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Abstract

For an integer $m \geq 3$, define

$$d = \gcd\big((m!)^{m!} - 1, (2m)!\big), \qquad t = \frac{d^d}{\gcd(d^d, d!)}.$$

Let a be the largest nonnegative integer such that $d^a \mid t$. We prove that

$$\frac{d}{\gcd\left(t/d^a, d\right)}$$

is the smallest prime greater than m. The construction has two stages: first d isolates exactly the primes in the interval $(m, 2m)$, and then the factorial $d!$ distinguishes the smallest of those primes by giving it the largest valuation.

Introduction

The problem considered here was mentioned to me by someone on IRC. Since the construction seemed mildly interesting, I took the liberty of thinking through why it works. The point of this note is not only to give a proof, but also to make the mechanism visible.

There are two separate filters in the formula. The first filter is

$$d = \gcd\big((m!)^{m!} - 1, (2m)!\big).$$

The factor $(2m)!$ prevents primes larger than $2m$ from appearing, while the congruence condition modulo $(m!)^{m!} - 1$ prevents primes at most m from appearing. What remains is precisely the set of primes strictly between m and $2m$.

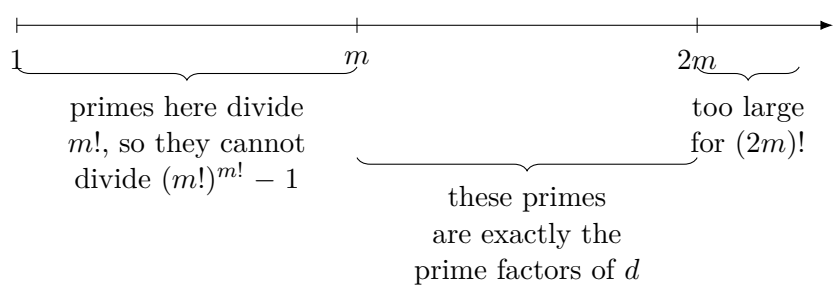

Figure 1: The first gcd filters primes according to their position relative to m.

The second filter is more subtle. Once d has been shown to be a squarefree product

$$d = p_1 p_2 \cdots p_k, \qquad m < p_1 < p_2 < \cdots < p_k < 2m,$$

the number d^d gives each prime factor p_i the same exponent d. Dividing by $\gcd(d^d, d!)$ subtracts the exponent with which p_i appears in $d!$. Smaller primes appear more often in $d!$, so the smallest prime p_1 is the one whose exponent is reduced the most. The definition of a then removes the common remaining power of d, and the final gcd detects which primes are still present. Exactly one prime has disappeared: p_1.

Statement

Theorem 1. *Let $m \geq 3$ be an integer. Define*

$$d = \gcd\big((m!)^{m!} - 1, (2m)!\big),$$

$$t = \frac{d^d}{\gcd(d^d, d!)}.$$

Let a be the largest nonnegative integer such that $d^a \mid t$. Then

$$\frac{d}{\gcd\left(\dfrac{t}{d^a}, d\right)}$$

is the smallest prime greater than m.

Proof

For a prime p and a positive integer N, write $\nu_p(N)$ for the exponent of p in the prime factorization of N.

We shall use Bertrand's postulate in the following form: for every integer $n > 1$, there is a prime p such that

$$n < p < 2n.$$

For completeness, an elementary proof of this fact is included in the appendix.

Lemma 1 (The first gcd)**.** *Let*

$$\mathcal{P}_m = \{p : \ p \text{ is prime and } m < p < 2m\}.$$

Then

$$d = \prod_{p \in \mathcal{P}_m} p.$$

In particular, d is squarefree and $d \neq 1$.

Proof. First suppose that a prime p divides d. Since $p \mid (2m)!$, we have $p \leq 2m$. On the other hand, $p \nmid m!$, because if $p \mid m!$, then

$$(m!)^{m!} - 1 \equiv -1 \pmod{p},$$

which is incompatible with $p \mid d$. Therefore $p > m$. Since $m \geq 3$, the endpoint $2m$ is composite, so $p \neq 2m$. Hence

$$m < p < 2m.$$

Thus every prime divisor of d lies in $\mathcal{P}_m$.

Conversely, let $p \in \mathcal{P}_m$. We prove that $p \mid d$. Since $p > m$, the prime p does not divide $m!$. Hence Fermat's little theorem gives

$$(m!)^{p-1} \equiv 1 \pmod{p}.$$

It remains to know that the exponent $m!$ is compatible with this congruence.

Except for the small case $(m, p) = (3, 5)$, one has

$$p - 1 \mid m!.$$

Indeed, write

$$p - 1 = 2q, \qquad q = \frac{p-1}{2}.$$

If $p = 5$, then $m = 3$ or $m = 4$. The case $m = 4$ gives $4 = p - 1 \mid 4!$, while the case $m = 3$ is the exceptional case just mentioned. If $p > 5$, then $q \geq 3$, and from $p < 2m$ we get $q < m$. Thus 2 and q are distinct factors occurring in $m!$, so

$$p - 1 = 2q \mid m!.$$

Therefore, outside the exceptional case,

$$(m!)^{m!} \equiv 1 \pmod{p}.$$

In the exceptional case $(m, p) = (3, 5)$, we have $m! = 6 \equiv 1 \pmod{5}$, so again

$$(m!)^{m!} \equiv 1 \pmod{p}.$$

Thus in every case $p \mid (m!)^{m!} - 1$.

Also $p \mid (2m)!$, since $p < 2m$. Hence $p \mid d$. Moreover, because $p > m$, we have $2p > 2m$, so p occurs exactly once in $(2m)!$. Consequently p occurs exactly once in d.

We have shown that the prime divisors of d are exactly the primes in $\mathcal{P}_m$, each with exponent 1. Hence

$$d = \prod_{p \in \mathcal{P}_m} p.$$

Finally, Bertrand's postulate ensures that $\mathcal{P}_m$ is nonempty, so $d \neq 1$. □

Now list the primes in $\mathcal{P}_m$ in increasing order:

$$\mathcal{P}_m = \{p_1, p_2, \ldots, p_k\}, \qquad p_1 < p_2 < \cdots < p_k.$$

By the lemma,

$$d = p_1 p_2 \cdots p_k.$$

Here $k \geq 1$.

Lemma 2 (The exponent filter)**.** *For each i, set*

$$f_i = \nu_{p_i}(d!).$$

Then

$$f_1 > f_2 > \cdots > f_k$$

whenever $k > 1$. Consequently, if

$$t = \frac{d^d}{\gcd(d^d, d!)},$$

then the largest nonnegative integer a such that $d^a \mid t$ is

$$a = d - f_1.$$

Proof. Since $d = p_1 p_2 \cdots p_k$ is squarefree, the exponent of p_i in d^d is exactly d. Also

$$f_i = \nu_{p_i}(d!) = \left\lfloor \frac{d}{p_i} \right\rfloor + \left\lfloor \frac{d}{p_i^2} \right\rfloor + \left\lfloor \frac{d}{p_i^3} \right\rfloor + \cdots .$$

Because $p_i > m \geq 3$, we have $p_i \geq 5$, and therefore

$$f_i \leq \frac{d}{p_i} + \frac{d}{p_i^2} + \frac{d}{p_i^3} + \cdots = \frac{d}{p_i - 1} < d.$$

Hence the exponent of p_i in $\gcd(d^d, d!)$ is exactly f_i, and so

$$t = \prod_{i=1}^{k} p_i^{d-f_i}.$$

It remains to compare the numbers f_i. If $i < j$, then $p_i < p_j$, so for every positive integer r,

$$\left\lfloor \frac{d}{p_i^r} \right\rfloor \geq \left\lfloor \frac{d}{p_j^r} \right\rfloor .$$

For $r = 1$, the inequality is strict, since d/p_i and d/p_j are integers and

$$\frac{d}{p_i} > \frac{d}{p_j}.$$

Thus $f_i > f_j$. In particular,

$$f_1 > f_2 > \cdots > f_k$$

when $k > 1$.

Let

$$e_i = d - f_i.$$

Then

$$t = \prod_{i=1}^{k} p_i^{e_i},$$

and the inequalities among the f_i reverse to give

$$e_1 < e_2 < \cdots < e_k$$

when $k > 1$. Since d is squarefree, $d^a \mid t$ exactly when every exponent e_i is at least a. Therefore the largest possible a is

$$a = \min_i e_i = e_1 = d - f_1.$$

The same formula is also valid when $k = 1$, where the minimum consists of the single exponent e_1. □

We can now finish the proof of the theorem. By the preceding lemma,

$$a = d - f_1.$$

Therefore

$$\frac{t}{d^a} = \frac{\prod_{i=1}^{k} p_i^{d-f_i}}{\prod_{i=1}^{k} p_i^{d-f_1}} = \prod_{i=1}^{k} p_i^{f_1 - f_i}.$$

The exponent of p_1 in this product is 0, while for every $i > 1$, the exponent of p_i is positive. Hence

$$\gcd\left(\frac{t}{d^a}, d\right) = p_2 p_3 \cdots p_k,$$

where the empty product is interpreted as 1 when $k = 1$. It follows that

$$\frac{d}{\gcd\left(\frac{t}{d^a}, d\right)} = \frac{p_1 p_2 \cdots p_k}{p_2 p_3 \cdots p_k} = p_1.$$

By construction, p_1 is the smallest prime in the interval $(m, 2m)$. Bertrand's postulate ensures that the least prime greater than m lies in this interval. Therefore p_1 is the smallest prime greater than m, as claimed.

Appendix: Bertrand's Postulate

For completeness, we include a standard elementary proof of the form of Bertrand's postulate used above.

Lemma 3. *For every real number $x \geq 1$,*

$$\prod_{p \leq x} p < 4^x,$$

where the product is over primes $p \leq x$.

Proof. It suffices to prove the claim for positive integers $x = N$, since replacing x by $\lfloor x \rfloor$ can only decrease the left-hand side.

We prove the integer case by strong induction on N. The cases $N = 1, 2$ are immediate. Suppose the result is known for all positive integers smaller than N.

First let $N = 2r$. The product of the primes p with $r < p \leq 2r$ divides the binomial coefficient $\binom{2r}{r}$. Hence

$$\prod_{p \leq 2r} p = \left(\prod_{p \leq r} p\right)\left(\prod_{r < p \leq 2r} p\right) < 4^r \binom{2r}{r} \leq 4^r \cdot 4^r = 4^{2r}.$$

Now let $N = 2r + 1$. The product of the primes p with $r + 1 < p \leq 2r + 1$ divides $\binom{2r+1}{r}$. Also

$$\binom{2r+1}{r} = \frac{2r+1}{r+1}\binom{2r}{r} < 2\binom{2r}{r}.$$

The central coefficient $\binom{2r}{r}$ is at most half of the sum of all coefficients in the $2r$-th row of Pascal's triangle; indeed, the two adjacent coefficients already have sum at least $\binom{2r}{r}$. Therefore

$$2\binom{2r}{r} \leq 4^r,$$

and hence $\binom{2r+1}{r} \leq 4^r$. Using the induction hypothesis at $r + 1 < 2r + 1$, we get

$$\prod_{p \leq 2r+1} p = \left(\prod_{p \leq r+1} p\right)\left(\prod_{r+1 < p \leq 2r+1} p\right) < 4^{r+1} \cdot 4^r = 4^{2r+1}.$$

This completes the induction. $\square$

Proposition 1 (Bertrand's postulate)**.** *For every integer $n > 1$, there is a prime p such that*

$$n < p < 2n.$$

Proof. Let

$$B_n = \binom{2n}{n}, \qquad P_n = \prod_{n<p\le 2n} p.$$

We first prove that $P_n > 1$ for $n \ge 468$. Since the central binomial coefficient is the largest coefficient in the expansion of $(1+1)^{2n}$, we have

$$B_n \ge \frac{4^n}{2n+1}.$$

Now factor B_n into primes. For a prime q, write $b_q = \nu_q(B_n)$. If $q \le \sqrt{2n}$, then each summand in

$$b_q = \sum_{r\ge 1} \left(\left\lfloor \frac{2n}{q^r} \right\rfloor - 2 \left\lfloor \frac{n}{q^r} \right\rfloor \right)$$

is at most 1, so $q^{b_q} \le 2n$. The total contribution from primes $q \le \sqrt{2n}$ is therefore at most $(2n)^{\sqrt{2n}}$.

If $\sqrt{2n} < q \le n$, then only the first summand can be nonzero. Moreover, for $2n/3 < q \le n$, one has

$$\left\lfloor \frac{2n}{q} \right\rfloor = 2, \qquad \left\lfloor \frac{n}{q} \right\rfloor = 1,$$

so such primes do not divide B_n. Hence

$$B_n \le (2n)^{\sqrt{2n}} \left(\prod_{q\le 2n/3} q \right) P_n.$$

By the previous lemma,

$$\prod_{q\le 2n/3} q < 4^{2n/3}.$$

Combining these inequalities gives

$$P_n > \frac{4^{n/3}}{(2n+1)(2n)^{\sqrt{2n}}}.$$

For $x \ge 468$, set

$$\phi(x) = \frac{x}{3}\log 4 - \log(2x+1) - \sqrt{2x}\log(2x).$$

Then

$$\phi'(x) = \frac{\log 4}{3} - \frac{2}{2x+1} - \frac{\log(2x)+2}{\sqrt{2x}}.$$

The last two terms decrease in magnitude as x increases, and a direct calculation gives

$$\phi'(468) > 0.17, \qquad \phi(468) > 0.1.$$

Thus $\phi(x) > 0$ for all $x \ge 468$. Equivalently,

$$\frac{4^{n/3}}{(2n+1)(2n)^{\sqrt{2n}}} > 1$$

for every integer $n \geq 468$. Hence $P_n > 1$, so there is a prime p with $n < p \leq 2n$. Since $2n$ is composite for $n > 1$, this prime satisfies $n < p < 2n$.

It remains only to check $2 \leq n < 468$. The following table covers all such n; in each row, the displayed number q is prime and satisfies $n < q < 2n$ throughout the indicated range.

q	range of n covered by $n < q < 2n$
3	$n = 2$
5	$3 \leq n \leq 4$
7	$5 \leq n \leq 6$
13	$7 \leq n \leq 12$
23	$13 \leq n \leq 22$
43	$23 \leq n \leq 42$
83	$43 \leq n \leq 82$
163	$83 \leq n \leq 162$
317	$163 \leq n \leq 316$
631	$317 \leq n \leq 467$

This proves Bertrand's postulate for every integer $n > 1$. $\square$